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LECTURE 1

Analog Integrated Circuits

Design of Analog CMOS Integrated Circuits

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

Textbook: Design of Analog CMOS Integrated Circuits — Behzad Razavi (UCLA)

Primary Textbook & Course Scope

- **Author:** Behzad Razavi — Professor, UCLA
- **Scope:** Covers analog CMOS design end-to-end — from MOS device physics and single-stage amplifiers to op-amps, bandgap references, switched-capacitor circuits, oscillators, and PLLs.
- **Role in this course:** Serves as the primary textbook; the 18-chapter table of contents (next slide) defines the course syllabus structure.

TABLE OF CONTENTS

18 Chapters, Six Thematic Groups

1 Foundations

1 Introduction to Analog Design

2 Basic MOS Device Physics

2 Basic Amplifiers

3 Single-Stage Amplifiers

4 Differential Amplifiers

5 Passive and Active Current Mirrors

6 Frequency Response of Amplifiers

3 System-Level Analysis

7 Noise

8 Feedback

9 Operational Amplifiers

10 Stability and Frequency Compensation

4 Specialized Circuits

11 Bandgap References

12 Intro. to Switched-Capacitor Circuits

13 Nonlinearity and Mismatch

5 Signal Generation

14 Oscillators

15 Phase-Locked Loops

6 Technology & Layout

16 Short-Channel Effects and Device Models

17 CMOS Processing Technology

18 Layout and Packaging

Why Is Analog Design Challenging?

Part 1 of 3

1. **Digital: a speed–power tradeoff**
   Digital circuits deal primarily with a speed–power tradeoff.

2. **Analog: a multi-dimensional tradeoff**
   Analog circuits deal with a multi-dimensional tradeoff of speed, power, gain, precision, supply, and more.

3. **High sensitivity to noise and interference**
   Due to speed and precision requirements, analog circuits are much more sensitive to noise, crosstalk, and other interferers.

Why Is Analog Design Challenging?

Part 2 of 3 (cont.)

4 **Sensitivity to second-order device effects**

Analog circuits are much more sensitive to second-order device effects.

5 **Limited design automation**

High-performance analog circuit design can rarely be automated — it typically requires hand-crafted design and layout.

6 **Modeling and simulation limitations**

Modeling and simulation of analog circuits are still problematic, requiring experience and intuition.

Why Is Analog Design Challenging?

Part 3 of 3 (cont.)

7 **Analog circuits on mainstream digital processes**
Economic forces require the development of analog circuits in mainstream digital processes.

8 **Analog–digital integration on a single substrate**
Economic forces push the integration of analog and digital functions onto a single substrate.

9 **Many levels of abstraction required**
Many levels of abstraction are required, from device physics to system-level architecture.